Safety Data Sheet

RENOLIN B 32 PLUS

1. Identification of substances

- **Product details :**

  Trade Name : RENOLIN B 32 PLUS

- **Application of the substance / the preparation:** Hydraulic fluid.

- **Manufacturer / Supplier :**

  FUCHS LUBRICANTS PTE LTD
  No 13A, Tech Park Crescent
  Tuas Teckpark
  Singapore 637843
  Tel : (65) 65588133
  Fax : (65) 68630603

- **Emergency information :**
  Tel : (65) 65588133

2. Hazards identification

- **Classification of the substance or mixture:**
- Classification according to #Directive 67/548/EEC or Directive 1999/45/EC#
  By handling of mineral oil products and chemical products no particular hazard is known when normal precautions (item 7) and personal protective equipment (item 8) are kept.
  The product is not classified according to Directive 67/548/EEC or Directive 1999/45/EC.
  - Void

- **Information concerning particular hazards for human and environment :**
  This product is water polluting, see item 12.
  This product does not have to be labelled due to the calculation procedure of the “General Classification guidelines for preparations of the EU” in the latest valid version.

- **Label elements :**
- **Labeling according to EU guidelines :**
  Observe the normal safety regulations when handling chemicals or mineral oil products.
  The product is not subject to identification regulations under EC Directives until 2008/1272/EC (1. CLP-ATP). Local regulations must be kept.

- **Other hazards :**
- **Results of PBT and vPvB assessment:**
  PBT :
  Not applicable.
  vPvB :
  Not applicable.

3. Composition / Data on components

- **Chemical characterization**

- **Description :**
  Mixture containing highly refined mineral base oils and additives.

- **Dangerous components :**

CAS No	Designation	%	Classification
4259-15-8	Zinc alkyldithiophosphate	<1.0	Xi, R41; N, R51/53; H318; H411
-	Butylated phenol	<0.5	-
101-02-0	Aryl Phosphite	<0.1	-
-	Calcium sulfonate	<0.1	-
74499-35-7	Substituted triazole	<0.05	-
64742-94-5	Petroleum naphtha	<0.05	-
-	Hydrocarbons, dearomatized	<0.02	-
-	N-Oleyl Diethanolamine	<0.01	-

4. First aid measures

- **Description of first aid measures**
- **General information:**
  Change clothes and shoes contaminated or soaked by the product.
  Never put rags contaminated by the product into cloth-pockets.
- **After inhalation:**
  Seek medical treatment in case of complaints.
- **After skin contact:**
  Instantly wash with water and soap and rinse thoroughly.
- **After eye contact:**
  Rinse opened eye for several minutes under running water.
- **After swallowing:**
  Seek medical treatment.

5. Fire fighting measures

- **Extinguishing media**
- **Suitable extinguishing agents:**
  C02, extinguishing powder or water jet. Fight larger fires with water jet or alcohol-resistant foam.
  Use fire fighting measure that suit the environment.
- **For safety reasons, unsuitable extinguishing agents:**
  Water with a full water jet.

- **Advice for firefighters**
- **Protective equipment:**
  In case of fire fighting: wear self containing breathing apparatus.

6. Accidental release measures

- **Personal precautions, protective equipment and emergency procedures:**
  Particular danger of slipping of leaked/ spilled product.

- **Environmental precautions:**
  Do not allow to enter drainage system, surface or ground water.
  Prevent from spreading (e.g. By binding or oil barriers).
  Do not allow to enter the ground/ soil.

- **Methods and material for containment and cleaning up:**
  Absorb with liquid-binding material (sand, diatomite, acid binders, universal binders, sawdust).
  Dispose of the material collected according to regulations.

- **Reference to other sanction:**
  No dangerous materials are released.

7. Handling and storage

- **Handling**
- **Precautions for safe handling:**
  Prevent formation of aerosols.
- **Information about protection against explosions and fires :**
  Do not heat up to temperature close to the flash point.

- **Conditions for safe storage, including any incompatibilities**
- **Storage**
- **Requirements to be met by storerooms and containers:**
  No special requirements.
- **Information about storage in one common storage facility:**
  Store away from foodstuffs.
- **Further information about storage conditions :**
  Local regulations concerning handling and storage of water polluting products have to be followed.

8. Exposure controls and personal protection

- **Additional information about design of technical systems :**
  No further data; see item 7.

- **Control parameters**
- **Components with critical values that require monitoring at the workplace :**
  The product does not contain any relevant quantities of materials with TLVs / WELs that have to be monitored at the workplace.

- **Additional information :**
  The lists that were valid during the compilation were used as basis.

- **Exposure controls**
- **Personal protective equipment:**
- **General protective and hygienic measures:**
- The usual precautionary measures should be adhered to in handling the chemicals and the mineral oil products**.**
- Avoid close or long term contact with the skin.
- Use skin protection cream for preventive skin protection.

- Do not carry cleaning cloths impregnated with the product in trouser pockets.
- **Breathing equipment:**
  Not required.
- **Protection of hands:**
  Protective gloves and protective skin cream.
- **Material of gloves :**
- The selection of the suitable gloves does not only depend on the material, but also on further marks of quality and varies from manufacturer to manufacturer. As the product is a preparation of several substances, the resistance of the glove material can not be calculated in advance and has therefore to be checked prior to the application.
- **Penetration time of glove material:**
  The exact break trough time has to be found out by the manufacturer of the protective gloves and has to be observed.
- **For the permanent contact gloves made of the following materials are suitable:**
  Nitrile rubber, NBR
  Neoprene gloves
- **Not suitable are gloves made of the following materials:**
  Butyl rubber, BR
  Natural rubber, NR
- **Eye protection:**
  Safety glasses recommended during refilling.
- **Body protection:**
  Protective work clothing**.**

9. Physical and chemical properties

- Form: Fluid.
- Colour: Light Brown.
- Smell: Characteristic
- Melting point / Melting range: Not determined
- Boiling point / Boiling range: Not determined
- Flash point: >200 Deg C
- Decomposition temperature: Not determined
- Self - inflammability: Product is not self-igniting.
- Danger of explosion: Product is not explosive.
- Density at 15 deg C: 0.86 g/cm^3
- Solubility in / Miscibility with water: Not miscible or difficult to mix.
- Viscosity at 40 deg C: 32 cSt

10. Stability and reactivity

- **Reactivity**
- **Chemical stability**
- **Thermal decomposition / conditions to be avoided :**
  No decomposition if used according to specifications.
- **Possibility of hazardous reactions :**
  No dangerous reactions known**.**
- **Hazardous decomposition products :**
  No dangerous decomposition products known.

11. Toxicological information

- **Information on toxicological effects**
- **Acute toxicity :**
- **Primary irritant effect :**
- **On the skin :**
  No irritant effect known
- **On the eye :**
  No irritant effect known
- **Sensitization :**
  No sensitizing effect known.

- **Additional toxicological information :**
  The product is not subject to classification according to the calculation method of the General EU Classification Guidelines for Preparations as issued in the latest version.
  When used and handled according to specifications, the product does not have any harmful effects according to our actual information.

12. Ecological information

- **Toxicity**
- **Aquatic toxicity:**
  No further relevant information available.
- **Behaviour in environmental systems :**
- **Mobility and bioaccumulation potential :**
  No data available.

- **Ecotoxical effects :**
- **Behaviour in sewage processing plants :**
  The product swims on the water surface.

- **Additional ecological information :**
- **General notes :**
  German Water Endangering Class 1 (self-assessment): slightly hazardous for water.
  Do not allow to reach ground water, water bodies or sewage system.

- **Results of PBT and vPvB assessment:**
  PBT: Not applicable
  vPvB: Not applicable

13. Disposal considerations

- **Waste treatment methods**
- **Recommendation :**
  Smaller quantities have to be disposed in line with local legislation.
  When storing used mineral oil products, ensure that the categories for waste oil and mixing instructions are observed.
  Delivery of waste oil only to officially authorized collectors.

- **European waste catalogue:**
  Mineral based non-chlorinated hydraulic oils

- **Uncleaned packagings :**
- **Recommendation :**
  Empty contaminated packagings thoroughly. They can be recycled after thorough and proper cleaning. Small one-way packaging have to be disposed according to the local regulations.

14. Transport information

UN-Number		
ADR	:	-
ADN	:	-
IMDG	:	-
IATA	:	-
UN Proper shipping name		
ADR	:	-
ADN	:	-
IMDG	:	-
IATA	:	-
Transport hazard classes		
ADR	:	-
Class	:	-
ADN / R Class	:	-
IMDG	:	-
Class	:	-
Packing group		
ADR	:	-
IMDG	:	-
IATA	:	-

- **Special precautions for user:** Not applicable**.**
- **Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code**
  Not applicable**.**
- **Transport/Additional information**
  No dangerous good acc. to dangerous goods/ transport directions**.**

15. Regulatory information

- **Safety, health and environmental regulations / legislation specific for the substance or mixture**

- **National regulations:**

- **German Water Hazard Class** :
  Water Endangering Class 1 (Self-assessment): slightly hazardous for water.
  Substances of very high concern (SVHC) according to REACH, Article 31 out of the recent list: none.

16. Other information

All ingredients are listed in the European Inventories.

These data are based on our present knowledge. However, they shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship. This data sheet is a safety data sheet according to 1907/2006/EC, article 31. For products which are not subject to classification according to EU lists this data sheet is made on a voluntary base.

- **Relevant phrases for the hazardous components in item 3:**
  H318 Causes serious eye damage.
  H411 Toxic to aquatic life with long lasting effects.
  R41 Risk of serious damage to eyes.
  R51/53 Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

- **Department issuing data specification sheet:**
  FUCHS LUBRICANTS PTE LTD
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